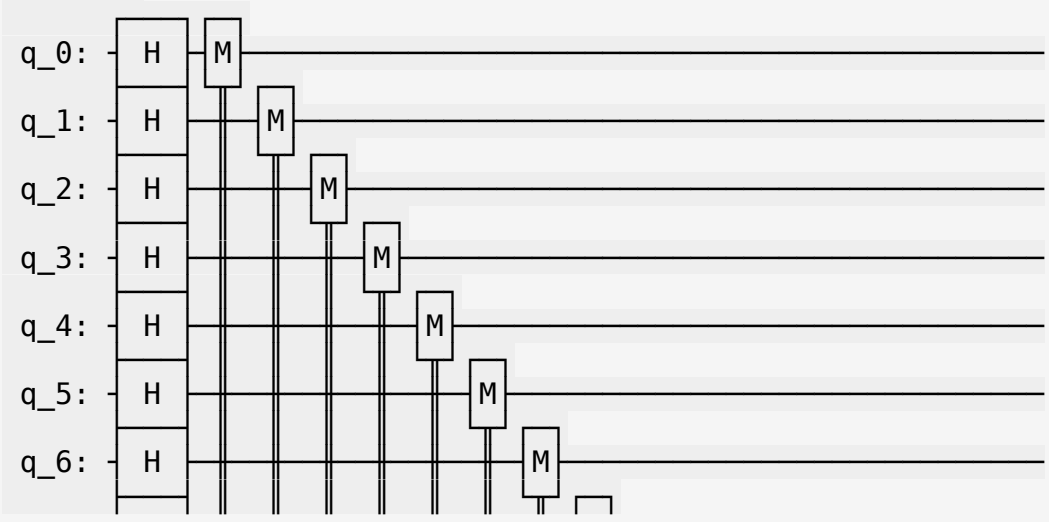
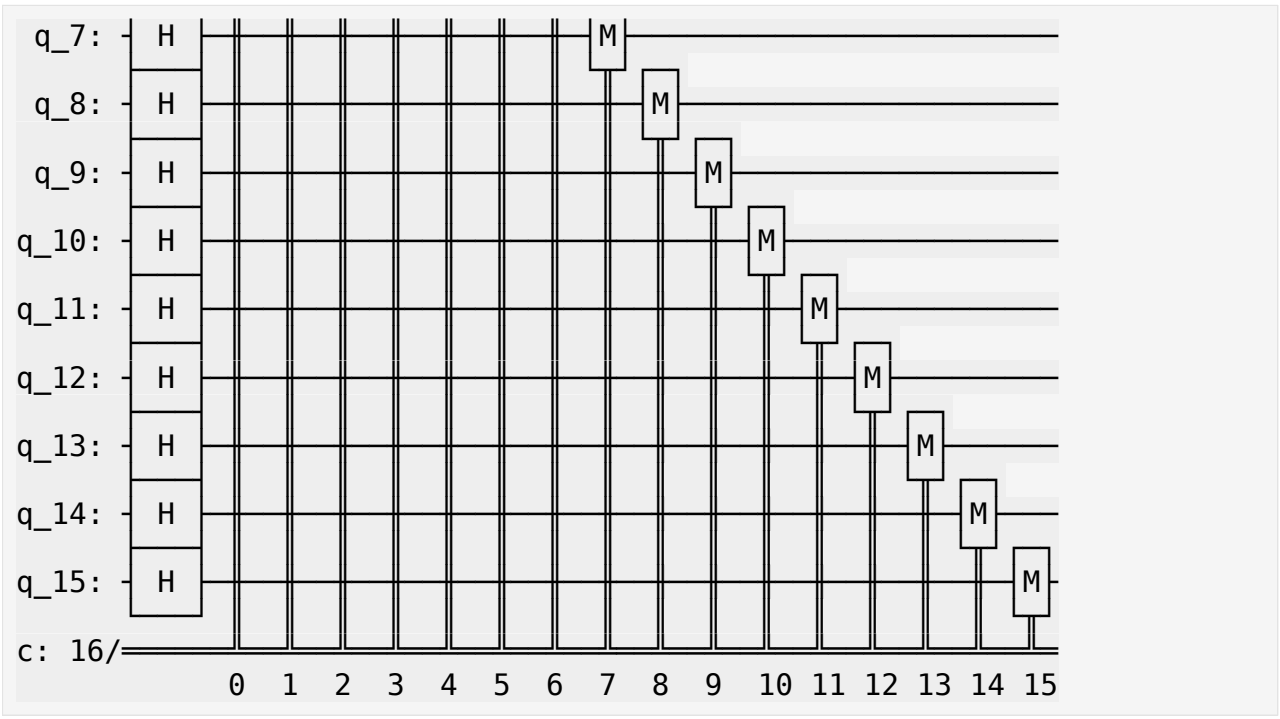
Practical 1

```python
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

# Create 16-qubit circuit
qc = QuantumCircuit(16, 16)

# Apply Hadamard gate to all qubits
qc.h(range(16))

# Measure all qubits
qc.measure(range(16), range(16))

# Run simulator
simulator = AerSimulator()
compiled = transpile(qc, simulator)
result = simulator.run(compiled, shots=1).result()

counts = result.get_counts()

random_number = list(counts.keys())[0]

print("16-Bit Quantum Random Number:")
print(random_number)

print("\nCircuit:")
print(qc.draw())
```

```
16-Bit Quantum Random Number:
0101010011100011

Circuit:
```




q_7:
q_8:
q_9:
q_10:
q_11:
q_12:
q_13:
q_14:
q_15:
c: 16/
H
M
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15